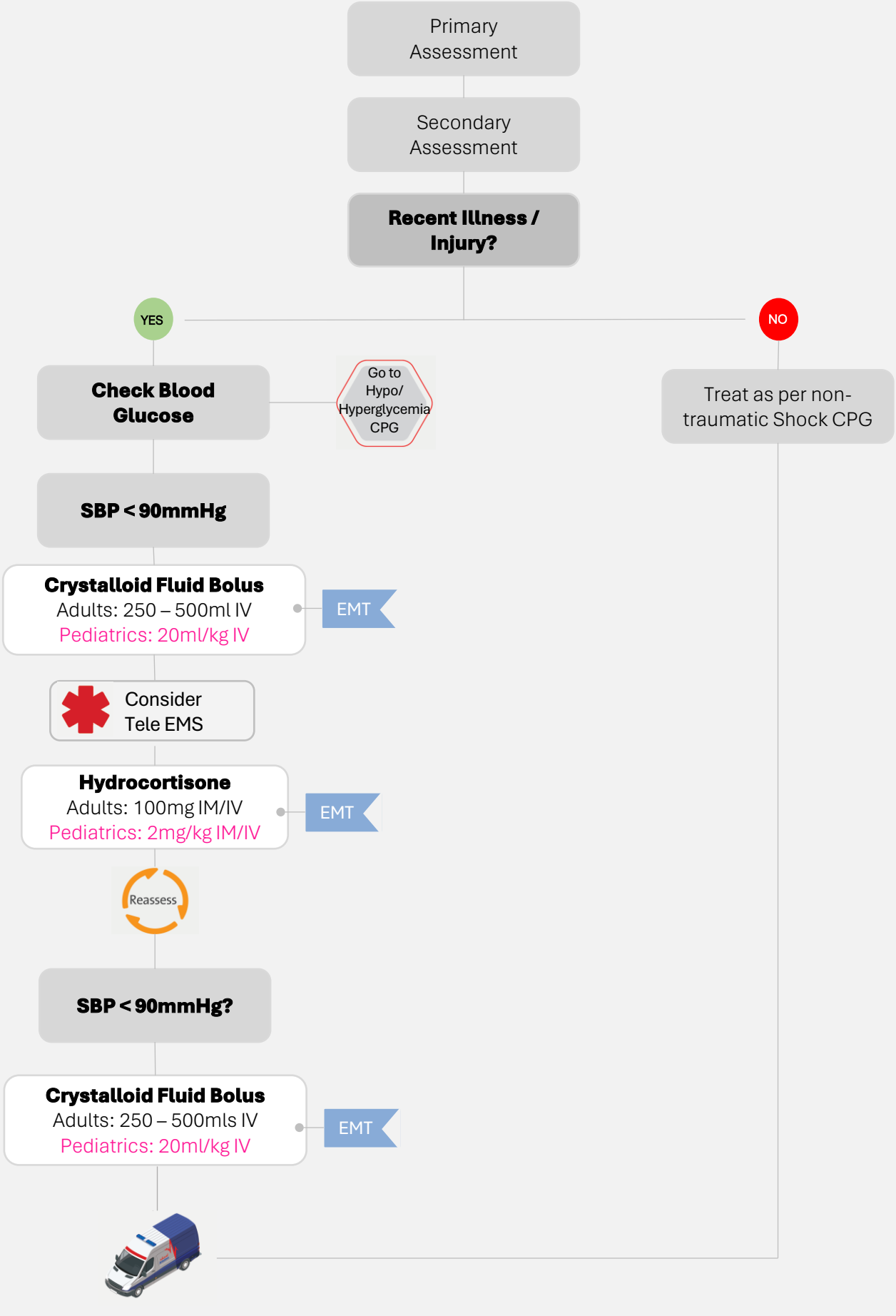
Adrenal Insufficiency



Patient Care Goals

1. Identification adrenal insufficiency (Adrenal Crisis/Addisonian Crisis).
2. Administration of corticosteroids and supportive care.

Patient Inclusion & Exclusion Criteria

Inclusion Criteria

- Patients presenting with symptoms suggestive of adrenal insufficiency or adrenal crisis, including:
  - Severe fatigue, weakness, or lethargy.
  - Nausea, vomiting, or abdominal pain.
  - Hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics).
  - Altered mental status or confusion.
  - History of adrenal insufficiency, chronic steroid use, or recent steroid withdrawal.
  - Known precipitating factors (e.g., infection, trauma, stress, or medication non-compliance).

Exclusion Criteria

- Symptoms clearly attributable to another cause (e.g., trauma-related shock, anaphylaxis, or cardiac arrest).
- Patients with no history or clinical suspicion of adrenal insufficiency.

Key Patient Assessment Points

1. **Patient History**
   - Confirm history of adrenal insufficiency (e.g., Addison's disease)
   - Inquire about chronic corticosteroid use or recent discontinuation.
   - Identify precipitating events (e.g., infection, fever, trauma, surgery, or stress).
   - Check for medical alert bracelets or documentation indicating adrenal insufficiency.
2. **Symptoms**
   - Assess for severe weakness, fatigue, or collapse.
   - Note nausea, vomiting, or diffuse abdominal pain without rigidity or rebound tenderness.
   - Evaluate for signs of hyperpigmentation (e.g., darkened skin creases, buccal mucosa) in primary adrenal insufficiency.
3. **Glucose Assessment**
   - Check random blood sugar (RBS) to identify hypoglycemia (<70 mg/dL).
4. **Determine Triage Category**
   - Determine Triage Category as per Patient Triage Categories CPG.

Patient Management & Interventions

1. **Primary and Secondary Survey**
   - Perform primary and secondary survey as per Universal Care CPG.
2. **Consider ALS back-up**
   - Consider the requirement for ALS back-up as per Universal Care CPG.
3. **Consider Red Flags**
   - Severe hypotension (SBP <90 mmHg in adults, or age-specific for pediatrics).
   - Persistent vomiting or inability to tolerate oral intake.
   - Altered mental status or unresponsiveness.
   - Suspected infection (e.g., fever >38°C, tachycardia).
   - Known adrenal insufficiency with acute deterioration.
4. **Evaluate for Signs of Shock or Instability**
   - Hypotension, tachycardia, altered mental status, or poor perfusion (e.g., cool, clammy skin, delayed capillary refill >2 seconds).
   - If shock is present, initiate fluid resuscitation and corticosteroid administration promptly.
5. **Establish IV Access**
6. **Fluid Therapy**
   - Administer Normal Saline or Ringers Lactate for signs of shock or dehydration:

        - Adults: 500-1000 mL bolus IV, reassess, repeat as needed.
        - Pediatrics: 20 mL/kg bolus IV, reassess, repeat as needed.
    - Monitor for fluid overload, especially in patients with comorbidities.
7. **Consider Hydrocortisone**
    - Indicated for Suspected Adrenal Crisis
        - Adults: 100 mg IV/IM for suspected adrenal crisis.
        - Pediatrics: 2 mg/kg IV/IM (max 100 mg).
8. **Consider Tele-EMS Consultation**
    - Consider calling Tele-EMS if the diagnosis of Adrenal Insufficiency is unclear or if advice on interventions is required.
9. **Patient Reassessment**
    - Frequently Reassess the patient as per Universal Care CPG.

Patient Safety Considerations

- Addisonian crisis is very difficult to diagnose in the ambulance setting, most patients will inform you of this condition – check for medical alert bracelets.

Clinical Insights & Additional Guidance

- An Addisonian crisis, also known as an acute adrenal crisis is relatively rare in the ambulance setting in Dubai.
- It is a life-threatening emergency caused by a sudden, severe deficiency of cortisol and often aldosterone due to adrenal insufficiency.
- Common Causes:
    - Medication non-compliance (most common)
    - Infection, trauma, surgery, dehydration.

Documentation

- Identification: Full name, age, and gender of the patient.
- Mechanism / Medical Complaint: History of adrenal insufficiency or chronic steroid use, presenting symptoms (e.g., weakness, vomiting, hypotension).
- Injuries / Information: Precipitating events (e.g., infection, stress), physical signs (e.g., hyperpigmentation), presence of alert bracelets.
- Signs: Vital signs, capillary refill, temperature, RBS level, mental status.
- Treatment:
    - IV access established.
    - Fluid bolus administration: type, volume, response.
    - Hydrocortisone given: dose, route, and time.
    - Telemedicine contacted and advice received (if applicable).
- Allergies: Known allergies, especially corticosteroids.
- Medications: Chronic steroid medications or immunosuppressants.
- Background: History of adrenal insufficiency, autoimmune disease, recent illness or surgery.
- Other:
    - Scene context, family or caregiver input, recent medication non-compliance.
    - Any deterioration en route or during handover to hospital.

Key Performance Indicators

1. Administer hydrocortisone of patients with suspected adrenal crisis.
2. Perform and document an RBS on all patients with suspected adrenal crisis.

References

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